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REVIEW

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Bridging AI and explainability in civil engineering: the Yin-Yang of predictive power and interpretability

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Abstract

Civil engineering relies on data from experiments or simulations to calibrate models that approximate system behaviors. This paper examines machine learning (ML) algorithms for AI-driven decision support in civil engineering, specifically construction engineering and management, where complex input–output relationships demand both predictive accuracy and interpretability. Explainable AI (XAI) is critical for safety and compliance-sensitive applications, ensuring transparency in AI decisions. The literature review identifies key XAI evaluation attributes—model type, explainability, perspective, and interpretability and assesses the Enhanced Model Tree (EMT), a novel method demonstrating strong potential for civil engineering applications compared to commonly applied ML algorithms. The study highlights the need to balance AI's predictive power with XAI's transparency, akin to the Yin–Yang philosophy: AI advances in efficiency and optimization, while XAI provides logical reasoning behind conclusions. Drawing on insights from the literature, the study proposes a tailored XAI assessment framework addressing civil engineering's unique needs—problem context, data constraints, and model explainability. By formalizing this synergy, the research fosters trust in AI systems, enabling safer and more socially responsible outcomes. The findings underscore XAI's role in bridging the gap between complex AI models and end-user accountability, ensuring AI's full potential is realized in the field.

Keywords Explainable AI (XAI), AI transparency, Causal reasoning, Sensitivity analysis, Feature relevance, AI in construction engineering, Data-driven engineering

1 Introduction

Experienced professionals often rely on intuition rather than systematic analysis when making complex decisions (Bakht & El-Diraby, 2015; Klein, 2004). While this method may offer expediency, it, at times, results in inconsistencies and errors. With the continuous evolution of technology, Artificial Intelligence (AI) has emerged as a revolutionary tool for augmenting human decision-making processes. Demonstrating prowess in scrutinizing extensive datasets characterized by high dimensionality, modern AI has the capacity to emulate human intuition by intricately transforming data points and vectors into elaborate matrices that mirror reality (Vincent, 2021). Despite these advances, AI systems still face significant limitations in logical reasoning—a critical requirement for engineering applications. Current AI models often operate without clear decision rules or mathematical frameworks, creating two major challenges: first, their internal causal mechanisms remain opaque, and second, engineering professionals consequently hesitate to fully trust AI-generated predictions (Shin, 2021).

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Researchers are increasingly turning to Explainable AI (XAI) to address these limitations, seeking to maintain AI's problem-solving advantages while making its decision processes understandable to users. This growing need for transparent AI systems is evident throughout civil engineering disciplines. Modern AI solutions must deliver not just accurate predictions, but clear explanations of how those predictions were generated. XAI elucidates the actions taken and their execution, underscores the pertinent information considered, and delineates the confidence levels associated with the resultant outputs. Such transparency builds trust and supports better decision-making (Gunning et al., 2019; Hasan & Lu, 2023a). For civil engineering applications, advancing XAI research remains crucial for successful AI adoption in professional practice.

The development of an AI model inherently entails a data-driven input-to-output mapping process founded on meticulously crafted machine learning algorithms. Historically, the challenges posed by high-dimensional data coupled with the necessity for substantial storage capacity often gave rise to computational inefficiencies, a conundrum commonly referred to as the "curse of dimensionality" (Thudumu et al., 2020). Dimensionality reduction serves as a mechanism to streamline intricate problems without resorting to overly simplistic solutions, thereby ensuring relevance and sufficiency in modeling endeavors. Progressions in cloud computing, storage facilities, and Graphics Processing Unit (GPU) technology have steadily dismantled these impediments, enabling AI systems to effortlessly handle vast, high-dimensional datasets (Do et al., 2020). Conversely, real-world data in construction engineering frequently harbors noise, inconsistencies, and gaps, culminating in an incomplete depiction of all pertinent dimensions of the problem under scrutiny (Waqar, 2024b). Despite the sophistication of AI algorithms, they remain susceptible to both systematic and stochastic errors (Hasan & Lu, 2023a). Consequently, akin to human intuition, the predictions posited by AI are not irrefutable; at times, AI veers into the realm of hallucination—propelling forth plausible yet counterfactual outcomes that diverge from physical actuality (Keane & Smyth, 2020).

At its core, the foundation of XAI is underpinned by the principles of explainability and interpretability (Love et al., 2023). These terms carry distinct meanings despite frequent conflation. Saeed and Omlin (2023) differentiate them as follows: explainability refers to generating audience-specific explanations that address particular needs, while interpretability measures how well these explanations align with the user's domain knowledge and make contextual sense. Essentially, explainability focuses on delivering relevant insights for expert validation, whereas interpretability evaluates whether those insights logically resonate with the user's knowledge and experience.

This research undertakes a specialized examination of the demands of construction engineering applications for XAI, delving into the elements of the problem, data, and model necessitating elucidation. The endeavor is geared towards formulizing XAI frameworks tailored for evaluating machine learning algorithms commonly applied within the domain of civil constructionengineering.

2 Overview of explainable AI

Since Rosenblatt's (1961) foundational work, the field of AI has undergone tremendous changes, and its revival in practical applications has been driven by machine learning (ML) (Marr, 2019). Although modern ML technology greatly improves predictive performance, its increasing complexity also reduces interpretability and blurs potential causal relationships in the data. This opacity is partly due to the traditional emphasis in computer science on functional outcomes rather than explanatory mechanisms—represented by the Turing test, which evaluates the performance of artificial intelligence based solely on behavioral outputs rather than reasoning processes (Levesque, 2017). In order to alleviate this limitation, interpretable XAI has become an important research field, attempting to reconcile the accuracy and interpretability of predictions. The concept was first formalized by Van Lent et al. (2004) in the context of game simulations, where systems were designed to articulate their internal decision-making logic. Later, Gunning et al. (2019) refined the definition within computing research, characterizing XAI as the development of secondary models capable of both explaining primary ML model behavior and maintaining sufficient predictive fidelity. Historically, XAI development was deprioritized during periods of rapid advancement in core AI performance metrics (Minh et al., 2022). However, as shown in Fig. 1, academic and industry interest has surged in recent years, with Google Trends data indicating an 80% increase in "explainable artificial intelligence" searches since 2023 compared to the previous decade. This trend is parallel to the wider adoption of artificial intelligence, and specifically reflects the increasing recognition that interpretability is a prerequisite for responsible deployment across technical domains.

2.1 Explainability and Interpretability

Within XAI frameworks, explainability constitutes a model's capacity to provide post hoc rationalizations of its operational logic (Khosravi et al., 2022). This characteristic exhibits inherent subjectivity, as its

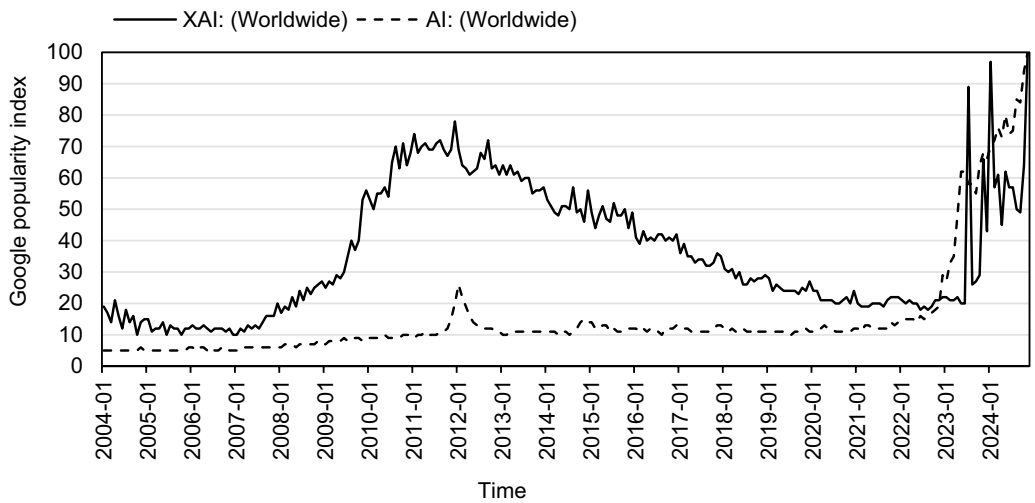

Fig. 1 Comparative trends in AI and explainable AI interest based on Google search data (normalized scale: 100 represents peak popularity)

effectiveness depends on end-users' ability to evaluate explanation plausibility and trustworthiness (Arrieta et al., 2020). For AI systems to demonstrate reliable explainability, three critical requirements must be satisfied (Love et al., 2023):

- Explainable Data: The data utilized to train the model must be clearly identified, including the rationale for their selection, the methods used to assess fairness, and the steps taken to mitigate bias.
- Explainable Predictions: The features activated and employed by the model to produce specific outputs must be transparent, and the reliability of the output should be comprehensible.
- Explainable Model: The individual components or layers of the model must be described, detailing their roles and contributions to generating the final output.

Interpretability in XAI refers to an AI model's built-in capacity to produce human-understandable outputs based on established knowledge (Fiok et al., 2022). Effective interpretability requires explanations that match users' cognitive frameworks, expertise, and potential biases (Gilpin et al., 2018). Interpretability emphasizes logical reasoning and transparent decision-making processes to make model predictions readily understandable. To enhance interpretability, Ribeiro et al. (2016) suggest providing users with visual or textual explanations that improve transparency and allow for inspection and replication. This is particularly important for ensuring fairness and ethical decision-making in AI systems. Transparency, as defined by Lipton (2018), refers to the degree to which humans can understand the internal mechanisms of predictive models, addressing the black-box nature of many AI systems. Belle and Papantonis (2021) identify three key dimensions of transparency:

- *Simulatability* Simulatability ensures that a model is simple enough for users to understand, accurately replicates the behavior it is designed to represent, and reliably reproduces predictions using input data and parameters, though even simple models can become opaque if complexity is not carefully managed.
- *Decomposability* Decomposability involves breaking down a model into its fundamental components—inputs, parameters, and computations—for individual analysis. However, this can be challenging when inputs are highly complex or poorly defined.
- *Algorithmic Transparency* At the algorithmic level, this dimension focuses on understanding the procedures a model employs to generate its output.

These three aspects of transparency—simulatability, decomposability, and algorithmic transparency—are essential for ensuring that predictive models remain interpretable despite their complexity. Beyond transparency, a model should also provide complete explanations, meaning they must allow stakeholders (developers, domain experts, users, and regulators) to reliably predict system behavior across various scenarios.

2.2 XAI model types

Explainability in XAI can be categorized into two types: post-hoc (Latin for "after this") and ante-hoc (Latin for "before this").

Post-hoc explainability provides local explanations after a decision is made, focusing on interpretability for specific outputs rather than the entire system. Key methods include:

- *LIME (Local Interpretable Model-Agnostic Explanations)* Developed by Ribeiro et al. (2016), LIME explains predictions by approximating the model locally with interpretable representations, such as linear models or decision trees.
- BETA (Black-Box Explanations through Transparent Approximations) (Lakkaraju et al., 2017) It balances fidelity and interpretability when explaining black-box classifiers.
- Pixel-wise Decomposition (Bach et al., 2015) It visualizes pixel-level contributions in neural networks and kernel-based classifiers, particularly useful in image analysis.

In contrast, ante-hoc explainability is built into the model architecture, ensuring inherent transparency. Examples include:

- Traditional interpretable models (e.g., linear regression, decision trees, fuzzy inference systems) (Guillaume, 2001).
- Generalized Additive Models (GAMs) (Caruana et al., 2015): Maintain interpretability while detecting patterns in high-dimensional data, such as medical datasets.
- Additive models for classifier decisions (Poulin et al., 2006): Enable visual explanations in bioinformatics applications using models like naïve Bayes and logistic regression.

Post-hoc methods excel at explaining opaque models, whereas ante-hoc approaches prioritize transparency by design, presenting trade-offs between flexibility and interpretability.

2.3 XAI Perspectives

Not all AI systems require explainability. This is particularly true when errors carry minimal risks or when the system has undergone rigorous validation and gained user trust—for instance, in advertising targeting or postal sorting applications (Adadi & Berrada, 2018). Nevertheless, determining when explanations become essential remains crucial. Saeed and Omlin's (2023) comprehensive literature review identifies multiple perspectives for assessing XAI necessity, which we examine below.

- *Regulatory Perspective* In domains where AI decisions carry substantial legal and ethical consequences, opaque systems raise serious accountability concerns. The EU's GDPR, for instance, explicitly grants individuals the right to demand explanations for algorithmic decisions affecting them—such as credit approvals (Goodman & Flaxman, 2017). However, without concrete technical solutions to produce meaningful explanations, these regulatory requirements risk becoming purely theoretical.
- *Scientific Perspective* While black-box models can effectively solve complex problems (Molnar et al., 2020), they essentially function as inscrutable knowledge repositories. XAI techniques serve a crucial role in unpacking this knowledge, making the underlying decision processes transparent and potentially leading to novel scientific insights across various fields.
- *Industrial Perspective* The business adoption of black-box AI faces dual challenges: regulatory constraints and user distrust. Many industries consequently prioritize interpretable models for compliance purposes, sometimes accepting reduced accuracy (Veiber et al., 2020). XAI helps navigate this interpretability-performance tradeoff, though implementing these solutions typically requires additional development resources and operational costs.
- *Developmental Perspective* Black-box systems are vulnerable to multiple failure modes including data biases, adversarial attacks, and overfitting. XAI provides essential tools for diagnosing these issues, improving system reliability while building user confidence. By revealing which features most influence decisions, XAI also enables more informed model comparisons and supports iterative refinement towards desired outcomes (Confalonieri et al., 2021).
- *End-User and Social Perspectives* Black-box systems can produce counterintuitive results, such as confidently recognizing imperceptibly altered inputs or failing on unrecognizable data. These behaviors raise concerns about reliability and fairness, especially when models reflect biases in training data (Guidotti et al. 2018). XAI clarifies decision-making processes, ensuring systems align with intended goals and fostering trust and fairness.

Across these diverse perspectives, one common theme emerges: explainability serves as a fundamental requirement for ensuring accountability and fairness, especially in AI applications that directly affect people's lives.

2.4 Explainability of models

AI research fundamentally seeks to build a solid theoretical framework for machine learning, one that supports autonomous systems capable of learning from experience without constant human oversight (Shahriari et al., 2015). However, realizing this vision presents multiple challenges: learning effectively from historical data, deriving meaningful insights, generalizing to new situations, handling high-dimensional data, and identifying key explanatory factors. A persistent challenge involves properly contextualizing data for specific applications. Both data quality and feature selection prove critical, with studies demonstrating that combining low-level features with high-level contextual information produces optimal results (Girshick et al., 2014). Consider these practical examples: In traffic monitoring systems, algorithms must differentiate between a single vehicle and a truck transporting multiple cars, requiring specialized features to properly identify transport trailers. Similarly, construction productivity models demand careful contextualization. When comparing welding tasks for bridge girder plates versus structural assemblies, accurate productivity measurement depends on precisely defined counting rules for labor hours in each specific scenario.

AI outputs can be categorized into point-value predictions and variance predictions, each requiring distinct sensitivity analysis techniques. These techniques are crucial for users to confidently and objectively rely on model predictions in engineering design, particularly when users make judgments and decisions subject to professional ethics and accountability. Take multiple linear regression (MLR) for example: the value and sign of the slope associated with each independent variable indicate the sensitivity of each predictor to the derived outcome. Similarly, in methods like multilayer perceptrons (MLP) or backpropagation neural networks (BPNN), first-order derivative-based techniques such as SHAP values quantify the sensitivity of inputs. It is noteworthy Wang et al. (2024) that formalized a method analogous to MLR for interpreting MLP sensitivity analysis; variance contribution factors were analytically defined for MLR to explain the sensitivity of each independent variable in regards to the variance predicted by MLR.

XAI models must address these challenges while ensuring that the data used is explainable and the model itself is interpretable. Whether employing post-hoc or ante-hoc approaches, the explainability and interpretability of XAI models must meet stakeholder expectations. Major stakeholders (such as AI users in the applied domain) determine the degree of transparency required, as their needs dictate the extent to which an AI model needs to be explained. Figure 2 presents a taxonomy for describing XAI models from various perspectives. This taxonomy will be applied in the subsequent sections to identify which aspects of construction engineering problems require explainability to develop a comprehensive XAI framework.

3 What needs to be explained

The expanding application of AI in civil engineering has spurred numerous reviews. However, these typically focus on specific sub-areas like building information modeling (BIM), automated construction manufacturing, and computer vision, offering a narrow view of AI's broader implementation within Construction Engineering and Management (CEM) (Hatami et al., 2019; Zhang et al., 2020; Zou et al., 2017). In their work, Pan and Zhang (2021) conducted both scientometric and qualitative analyses to elucidate the current state of AI in CEM and outline future research trends. Furthermore, the growing interest in explainable AI (XAI) in construction engineering has prompted narrative reviews and taxonomies to enhance understanding and application in this field (Love et al., 2023). These reviews have contributed to developing a "Content, Context, Process, and Outcome" evaluation framework that aids construction organizations in leveraging XAI. However, existing literature often overlooks the intricate relationship between specific problem aspects and XAI, suggesting that the definition of an explainable AI solution requires further clarification and formalization. The subsequent sections will describe how to define an XAI algorithm tailored to the unique needs of civil engineering, particularly in construction engineering and management.

A conceptual framework is presented in Fig. 3. Any XAI algorithm can be divided into three fundamental phases: training, testing, and deployment. In the following sections, the explainability requirements essential for rendering effective XAI are described.

3.1 Model training

3.1.1 Configuring

In the model learning phase, before training begins, the data must be preprocessed to ensure suitability for AI model training. Real-world data used in AI modeling often contains noise due to inconsistencies or errors in data collection, which can affect prediction model performance. In practice, construction plans, such as cost budgets for specific scopes of work, are presented as baselines for field execution. To ensure successful implementation, planners (or estimators) must explain and communicate: (1) the reasoning process of the AI model in relating input factors to predicted outputs,

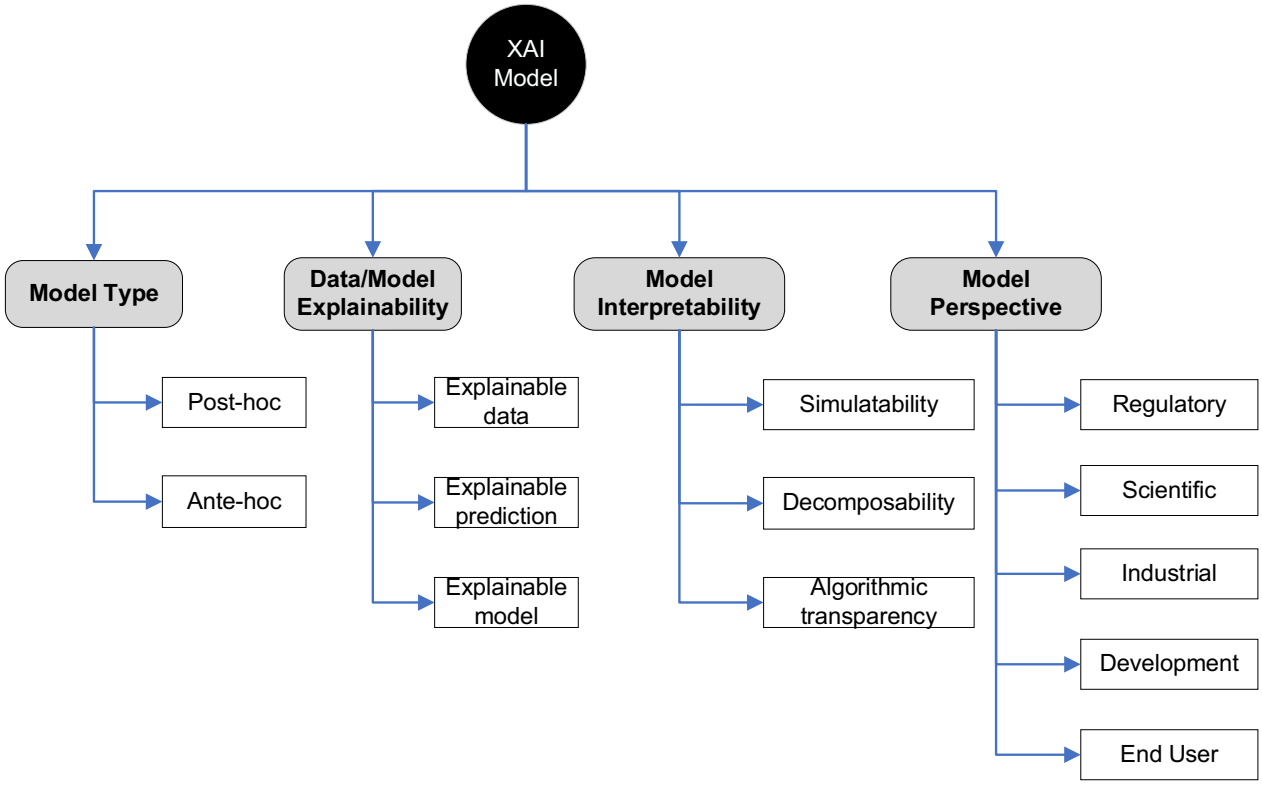

Fig. 2 Conceptual taxonomy of XAI model

(2) the extent of noise in the learning data and its potential effect on prediction performance, and (3) the trustworthiness of specific predictions relative to the model's overall accuracy. These aspects underscore the critical importance of explainability in AI models to ensure their acceptance and effective use in construction engineering and management. AI models for engineering must provide explainability on multiple fronts: the relevance and sensitivity of input factors, the distinction between signal and noise, and the impact of training data changes on prediction accuracy and uncertainty. Developing techniques to address these aspects ensures a more robust deployment of AI solutions tailored to specific engineering applications.

Input features and their predictive role An AI model's function $y = F(x)$ relies on input vector X to predict target variable Y. While X components should ideally be observable and measurable, their predictive relevance varies significantly. Some inputs reflect physical laws or measurable phenomena, while others may originate from empirical observations or even misconceptions—creating potentially "fictitious" factors. Identifying genuinely predictive features requires rigorous statistical methods like stepwise regression (Mohsenijam et al., 2016) and robust correlation analyses.

Challenges with missing and irrelevant input factors Critical input factors are sometimes omitted due to measurement challenges or knowledge gaps, leading to poorly defined problems and models that fail to converge. Paradoxically, a model might achieve good predictions while missing important inputs suggesting dataset-specific invariance. However, such models often fail when applied more broadly where the omitted factors become relevant. This highlights the risks of overfitting to specific datasets at the expense of generalizable solutions.

Distinguishing signals from noise in training data Engineering datasets typically contain substantial noise, requiring models to clearly separate meaningful patterns from random variations. During model configuration, it's essential to explicitly flag portions of data deemed noisy, as intelligence derived from these segments may be unreliable or misleading. In the configuration phase, the presence of noise is confirmed, and the impact of the noise on the prediction must be realized.

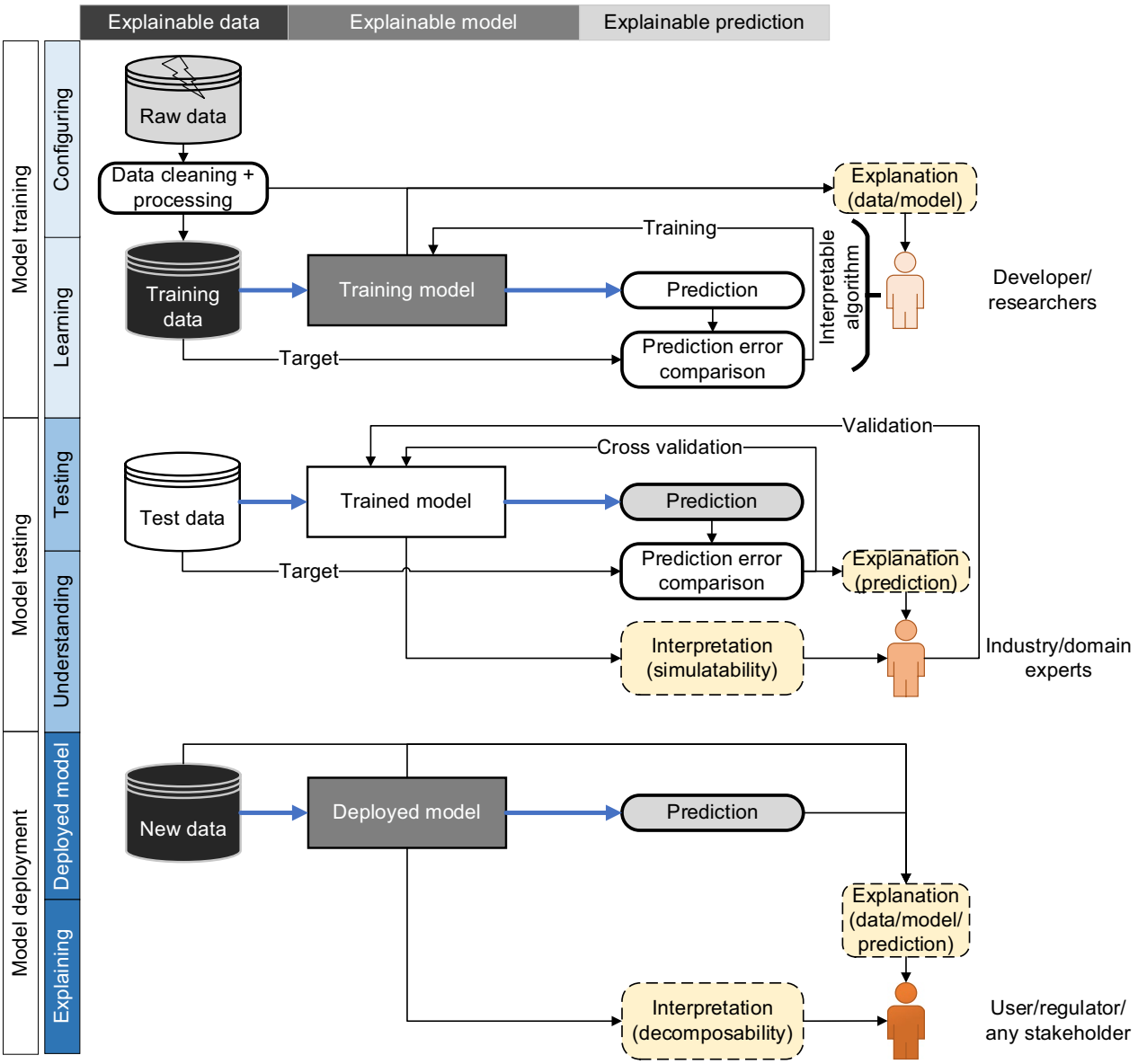

Fig. 3 Architecture of explainable artificial intelligence (XAI) in Civil Engineering applications

3.1.2 Learning

Besides configuring the data used for training the model, the structure of the model, including individual components or layers, should be clearly defined, detailing their roles and contributions to the final output. Most ML algorithms operate recursively, meaning that a sequence of steps is repeated until specific stopping criteria are satisfied. The learning phase involves fine-tuning model settings, known as hyperparameters, which are not directly learned from the data but are devised by heuristics or trial and error. These settings may include the learning rate, the number of layers in a neural network, or the number of trees in a random forest. Hyperparameter tuning is commonly conducted using methods like grid search or random search to identify the combination that yields the best performance. It is recommended that the iterative stages of the learning algorithm, including defining hyperparameters, fine-tuning them, and deciding when to halt training, are clearly disclosed to the modeler.

3.2 Model testing

3.2.1 Testing

The testing phase evaluates model performance using a holdout dataset that was withheld during

training. This test set must satisfy two key criteria: (1) representativeness of real-world problem conditions, and (2) sufficient independence to properly assess learned patterns and potential biases. The fundamental objective is to validate the model's generalization capability—its capacity to maintain performance beyond the training data distribution. Evaluation requires systematic measurement using quantitative metrics like accuracy and precision. More critically, results analysis should specifically check for overfitting or underfitting. Overfitting occurs when a model is too closely fitted to the training data, capturing noise instead of the underlying pattern, thereby performing poorly on new or unseen instances. Underfitting, on the other hand, happens when a model is too simplistic to capture the complexity of the data it was trained on. Manyfold cross-validation technique is a good way to check the model overfitting and underfitting issues, at least when available data is scarce.

3.2.2 Understanding

In the context of explainable AI performance, it is crucial for the model's explanations to be in harmony with the domain expert's internal mental framework consisting of experience, prior knowledge, and intuition. The model's quantitative validation estimates must align with the domain expert's implicit or tacit knowledge. Such validation from the human expert perspective confirms the simulatability of the model itself. A commonly overlooked aspect in the literature is the selection and identification of the expert who interacts with the XAI. This research intends the experts to be professional engineers who have both formal education and practical experience in engineering.

A critical research focus involves analyzing how training data affects prediction uncertainty. Consider this specific case: When modeling wall panel productivity, integrating additional BIM data from recent projects may alter the slope coefficients in MLR analyses, consequently changing output predictions. As Hasan and Lu (2023a) demonstrate, precisely measuring how such data variations affect prediction uncertainty is vital for ensuring AI reliability. These rigorous data handling and explanation practices make XAI systems truly transparent and practical for engineering professionals and stakeholders.

3.3 Model deployment

The deployment of AI models demands a comprehensive approach encompassing three key aspects: system integration, operational monitoring, and compliance management. First, models must be properly integrated into the target environment, typically through encapsulation as API services or containerized applications. This packaging should facilitate seamless interaction with existing systems while maintaining robust data exchange protocols.

Ongoing monitoring forms the second critical component, requiring continuous performance tracking to identify potential model drift and schedule necessary updates as data patterns evolve. The monitoring system should generate alerts when prediction quality metrics fall below established thresholds.

From a compliance perspective, deployed models must adhere to stringent data privacy standards and ethical guidelines while providing complete transparency. This includes detailed documentation of input–output relationships, accuracy estimates, and validation methodologies. For engineering applications, these requirements carry particular significance as professionals remain ultimately accountable for decisions informed by model outputs. There needs to be a quick check mechanism to indicate the validity of the model prediction or to verify the model's prediction validation process.

4 Enhanced model tree (EMT): ideal XAI solution

Model Tree algorithms implement classification through a "divide and conquer" approach (Solomatine and Xue, 2004). The M5P algorithm, first developed by Quinlan (1992) and subsequently enhanced by Wang and Witten (1997), operates by recursively splitting the dataset into subsets via binary decision rules, generating a hierarchical tree structure. Each internal node represents a decision based on a feature value, while each leaf node contains a MLR model tailored to its specific subset. Figure 4 illustrates this concept. The left panel shows a single linear regression line fitted to a heterogeneous dataset, resulting in low predictive accuracy. In contrast, the right panel demonstrates how partitioning the dataset into three subsets, each defined by the model tree's branching rules, allows for fitting distinct linear regression models. This partitioning leads to improved predictive performance by effectively capturing patterns from local data.

The traditional Model Tree algorithm was enhanced by Hasan and Lu (2023a), with the key advancement being the integration of binary data classification and variance analysis to complement point value predictions. Incorporating variance analysis is critical for decision-making in safety-sensitive applications, such as concrete strength prediction (Hasan & Lu, 2023b), structural health monitoring for damage detection (Salkhordeh et al., 2021), and geotechnical assessments like ground settlement and landslide prediction (Gao, 2024). Moreover, variance analysis also adds significant value in non-safety–critical applications, such as project cost

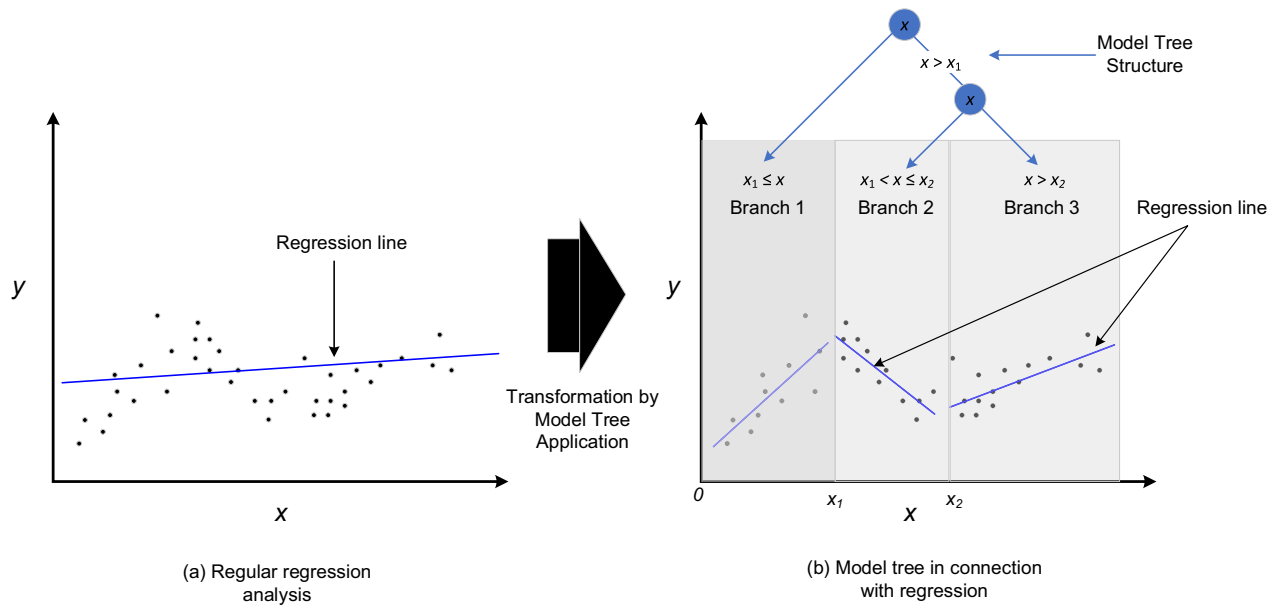

Fig. 4 Illustration of improvement of regression analysis with model tree application

risk analysis (Hasan & Lu, 2023a), where understanding uncertainty is essential for informed and reliable decision-making. The subsequent sections detail the EMT algorithm's architecture and evaluate its alignment with core XAI principles.

4.1 EMT architecture

The recursive model tree algorithm operates by selecting both the most informative feature and its optimal split value to partition the data into subsets exhibiting homogeneous target behavior. This optimal split is determined through maximization of error reduction—quantified by standard deviation minimization—using the Standard Deviation Reduction (SDR) criterion (Wang & Witten, 1997), as formally expressed in Eq. 1.

$$SDR = STD(T) - \sum_i \frac{T_i}{|T|} STD(T_i) \tag{1}$$

$$STD = \sqrt{\frac{\sum_j (y_j - \bar{y})}{n}} \tag{2}$$

Here, T is the data at a given node, T_i denotes the resulting subsets after splitting, and STD (Eq. 2) calculates the standard deviation. A split at value a on variable x_i yields two groups: one with $x_i \leq a$, and another with $x_i > a$. Splitting continues until further division offers negligible error reduction or too few instances remain. Finally, an MLR model is fitted at each leaf using its data subset. Initial data classification performance is determined using the coefficient of determinant (R^2).

The EMT extends this by incorporating variance analysis into the MLR models at each branch, thereby enabling prediction variance estimation. This variance analysis is implemented through sampling data N times to generate N MLR models. Also, it is done for all M branches of the model tree model. For each coefficient, β_n, the mean and standard deviation (μ, σ) are calculated. The prediction regression model is given in Eq. 3:

$$Y(\mu_Y, \sigma_Y) = \beta_0(\mu_0, \sigma_0) + \beta_1(\mu_1, \sigma_1)x_1 + \cdots + \beta_n(\mu_n, \sigma_n)x_n \tag{3}$$

The error propagation as presented by Eq. 4 then can be used in connection with the regression model to derive the variance estimate model in association with the MLR prediction model. Here, C_Y is the covariance of the prediction, $J_{\beta Y}$ is the Jacobian of the MLR model and C_β is the covariance of the MLR model coefficients.

$${\sigma_Y}^2 = C_Y = J_{\beta Y} C_\beta J_{Y\beta}^T \tag{4}$$

Thus, each branch of the EMT has two equations, one to make the point prediction and the other to estimate the associated variance of the prediction. To assess model reliability, a variance index $\alpha = \frac{\sigma_P}{P}$ is used, where σ_P is the standard deviation of mean point prediction P. If $\alpha < \alpha_c$, the model is accepted; otherwise, it is rejected. The threshold α_c depends on domain-specific standards (e.g., 95% in construction, 99% in medical fields).

The architecture of the enhanced model tree algorithm is presented in Fig. 5. Elaborated step-by-step development of the enhanced model tree algorithm can be found in Hasan (2023).

4.2 Evaluation of EMT across different XAI domains

Compared to ANNs, model trees represent a logic-driven machine learning approach that inherently maintains transparency in modeling input–output relationships. The enhanced model tree (EMT) offers two distinct advantages over conventional machine learning methods: (1) it provides analytical prediction of variance estimates in conjunction with point-value outputs, and (2) maintains explainability through clear prediction logic. This structural transparency is critical for establishing practitioner trust and facilitating real-world AI implementation. Table 1 systematically evaluates the EMT algorithm within the XAI framework, examining its suitability across multiple assessment dimensions.

4.3 Evaluation of commonly applied machine learning algorithms

Building upon the previous evaluation of the EMT algorithm's alignment with various XAI domain dimensions, this section presents a comparative assessment of conventional machine learning approaches used in civil engineering applications. The analysis specifically examines these alternative methods in contrast to the EMT algorithm, with particular focus on their respective interpretability and explainability characteristics.

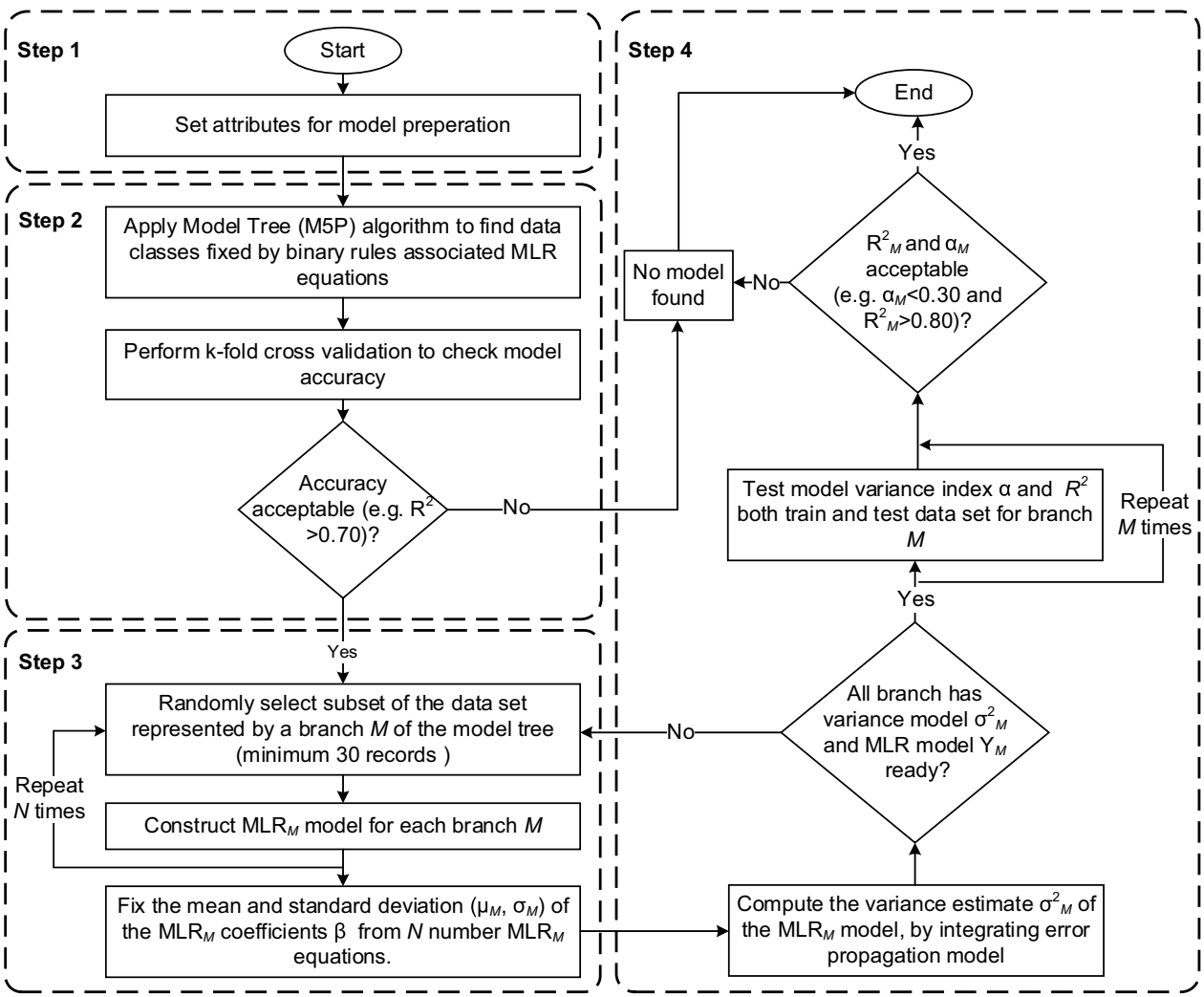

Fig. 5 Enhanced model tree algorithm architecture

Table 1 Evaluation of the EMT algorithm across different XAI domain dimensions

Attribute	Instances for EMT	Remarks	Explanatory notes
Model Type	Ante-hoc	Yes	The model maintains inherent explainability during its construction phase, ensuring full transparency throughout the development process The entire model architecture remains explainable, rather than just individual outputs
	Post-hoc	No	The model does not rely on after-the-fact explanations
Explainability	Data	Yes	To confirm the performance of the model tree algorithm, it's essential to ensure the dataset is free from noise and missing values. While the algorithm automatically identifies relevant attributes for rule generation and MLR development, preliminary dimensionality reduction during preprocessing can enhance results
	Model	Yes	EMT prediction models are essentially MLR equations constrained by binary rules on specific attributes of the dataset, following a tree-like hierarchical structure The MLR model's corresponding rule sets are fixed, thus making it transparent and easily comprehensible
	Prediction	Yes	The EMT algorithm provides variance estimates for point predictions, serving as a reliability indicator for users
Model Perspective	Regulatory	Yes	Since the EMT algorithm is based on MLR equations, the input data and processing mechanisms are easily verifiable, allowing outputs to be readily traced back to their corresponding inputs
	Scientific	Yes	Like any other machine learning model, the EMT supports scientific discoveries by revealing hidden patterns in the relationships between inputs and outputs
	Developmental		The EMT specifically provides advancements in identifying small data subsets and models that significantly impact prediction variance. This capability allows for pinpointing areas of the process that require improvement or additional data for model refinement
	End-user	Yes	The model is easy to use; it can be coded in a computing environment or even used with pen and paper
	Industry	Yes	The model can solve practical industry problems, as demonstrated in recent case studies (Hasan & Lu, 2023a, 2023b), and offers transparency in the input–output relationship. The model's performance acceptance threshold, namely the "variance index", is set by the practitioner
Interpretability	Simulatability	Yes	The produced model by the EMT algorithm is simple enough for users to understand the inputs that produce the predictions and results are reproducible
	Decomposability	Yes	The resulting models from the model tree algorithm are distinguished by the inputs, models, and outputs
	Algorithmic transparency	Yes	The EMT algorithm recursively partitions the dataset into subsets using binary decision rules and represents the relationships between inputs and outputs using MLR equations, along with an ad hoc error propagation model that operates atop the established MLR model. Model parameters such as the variance index and the number of instances for each branch are predefined

4.3.1 Linear regression:

- *Data* The model performs optimally with a minimal set of mutually independent input variables exhibiting negligible collinearity. Residual analysis and coefficient p-values serve to evaluate input relevance.
- *Model* Utilizing ordinary least squares (OLS), the model requires verification of three key assumptions through diagnostic plots: linearity (residual plots), homoscedasticity (residual vs. fitted values), and normality of residuals (Q-Q plots). Model variables and coefficients remain fixed upon finalization.
- *Prediction* Generates point predictions with uncertainty quantified via residual standard error or standard error of prediction. Note that predictive accuracy may be compromised by outlier sensitivity.
- *Simulatability* Features an additive structure where each input is weighted by its coefficient, ensuring straightforward interpretation with small feature sets.
- *Decomposability* Explicit coefficient values and intercept term enable clear interpretation of individual variable effects (ceteris paribus) (Strejc, 1981).
- *Algorithmic Transparency* Coefficients are found by minimizing squared errors, often solvable analytically (Nimon & Oswald, 2013). Gradient methods, if used, are reproducible (Ancona et al., 2019).
- *Model Type* Classified as an ante hoc model.

4.3.2 K-Nearest neighbors (KNN):

- *Data* Relies on storing training examples. Quality, relevance, and feature scaling are critical. No explicit training phase, but choosing salient features and removing irrelevant ones is beneficial. Cross-validation determines the optimal *k*.
- *Model* Instance-based—no parameters are learned. Distances (often Euclidean) are computed as basis for making inferences. While efficient for small datasets, computational costs escalate with data growth.
- *Prediction* Predictions derive from aggregating (voting/averaging) the k-nearest neighbors to each query point, with performance contingent on distance metric and k-value selection.
- *Simulatability* Conceptually straightforward to follow (distance calculation, neighbor selection, aggregation), but large datasets make manual simulation impractical (Zhang, 2021).
- *Decomposability* No global formula or coefficients—each prediction depends on local neighborhood data, offering partial interpretability.
- *Algorithmic Transparency* Steps—distance computation, neighbor selection, aggregation—are clear. No closed-form solution, relying on distance evaluations for each inference.
- *Model Type* An ante hoc "lazy learner" with instance-based interpretability.

4.3.3 Bayesian model:

- *Data* Requires well-defined data and prior domain knowledge. Quantitative and qualitative features should be carefully handled to align with prior distributions/likelihood functions. Missing or outlier data can be included in the probabilistic framework.
- *Model* Specifies prior distributions over parameters, a likelihood function for data generation, and yields a posterior distribution. Beliefs about parameters are updated with observed evidence. Markov chain Monte Carlo simulation (MCMC) or Variational Inference approximates this posterior if no closed-form solution exists.
- *Prediction* Derived from the posterior predictive distribution, integrating uncertainty in parameters and data. Predictive intervals highlight confidence around predictions.
- *Simulatability* The concept of updating beliefs is easy to grasp, but performing exact calculations, especially with MCMC, can be complex. Simple conjugate priors allow for manual calculations, while complex models require computational tools.
- *Decomposability* Emphasizes the distributions of parameters rather than individual coefficients, making it less straightforward to assign per-feature effects. However, techniques such as partial dependence plots or posterior summaries can aid in understanding these effects (Gelman et al., 2013).
- *Algorithmic Transparency* Despite the logical clarity in separating priors, likelihoods, and posterior updates, iterative methods like MCMC can obscure the process if not closely monitored. Each step is rooted in probability theory, ensuring conceptual clarity throughout the modeling process.
- *Model Type* The probabilistic modeling paradigm is often considered a priori, with the structure (prior, likelihood) specified at the outset for interpretability and quantifying uncertainty (Ali et al., 2023).

4.3.4 Decision tree model:

- *Data* Suitable for handling both numeric and categorical data, requiring minimal preprocessing. Handling missing values is necessary. However, including overly correlated or irrelevant features may result in complex or overfitted trees.
- *Model* Constructs a tree by recursively splitting the data based on specific criteria, such as variance reduction, to isolate classes or regression outcomes. Nodes and branches represent key decision points, while leaves provide the final labels or numeric predictions.
- *Prediction* For classification, new data propagates from the root to a leaf for the predicted label. For regression, leaves often store the mean outcome of training points within each node.
- *Simulatability* Straightforward for smaller trees—one can trace "if–then" paths manually. Deep or large trees can become unwieldy.
- *Decomposability* The tree partitions feature space into distinct regions but does not produce additive contributions for each variable. It is interpretable in a rule-based sense, though not expressed as a single global formula.
- *Algorithmic Transparency* Splits are chosen to reduce impurity/error. Each choice can be visualized or observed. Hyperparameters (e.g., pruning depth) need careful tuning and explanation (Mahbooba et al., 2021).
- *Model Type* Decision trees are typically considered ante hoc models since their structure can be visualized and understood before deployment. However, the interpretability may decrease with very large trees.

4.3.5 Support vector machine (SVM):

- *Data* Usually assumes numerical inputs. Proper scaling (normalization) is crucial. Outliers or imbalance can influence margin-based decisions. Feature selection is not mandatory but can improve performance.
- *Model* Seeks an optimal separating hyperplane (for linear SVM) or boundary (for nonlinear kernels), maximizing the margin. Support vectors—data points near the boundary—are vital.
- *Prediction* For classification, determines which side of the boundary a new point falls on; for regression, approximates a continuous value. Kernels can map data into higher dimensions to capture complex patterns.
- *Simulatability* Linear SVMs are somewhat simulative (similar to linear models), but nonlinear kernels complicate manual tracing of predictions (Kiely & Gielen, 2004).
- *Decomposability* No feature-level breakdown. The kernel function and support vectors spread influence across the dataset. Partial dependence plots can offer insights, but no direct global formula exists.
- *Algorithmic Transparency* At a high level, margin maximization and kernel transformations are conceptually straightforward. However, kernel parameters (e.g., gamma for RBF) can be opaque (Kim et al., 2020).
- *Model Type* Often seen as black-box or post-hoc explainable, especially with complex kernels. Trades interpretability for predictive power in certain contexts.

4.3.6 Artificial neural network (ANN):

- *Data* Typically needs large datasets to capture complex patterns. Features should be normalized for stable training. Missing or noisy data, if not managed, can lead to overfitting.
- *Model* Composed of interconnected layers of neurons with weights and biases. Uses activation functions to enable nonlinear transformations. Training employs backpropagation and gradient-based optimization to minimize a chosen loss function.
- *Prediction* Feeds input through hidden layers to an output layer, producing class probabilities (classification) or continuous values (regression). Capable of modeling highly nonlinear relationships with accuracy.
- *Simulatability* While neuron operations are simple, the vast number of parameters makes manual simulation impractical, lacking the straightforward rule sets of trees or linear models (Fan et al., 2021).
- *Decomposability* Does not naturally decompose predictions by feature. Each neuron aggregates inputs in complex ways, obscuring direct variable-specific effects. Techniques like SHAP, LIME, or layer-wise relevance propagation add partial explainability.
- *Algorithmic Transparency* Forward/backward passes are well-defined, but weight matrices and embeddings are not intrinsically interpretable. Tuning hyperparameters (layer depth, neuron count, learning rate) can be non-intuitive (Greif, 2022).
- *Model Type* Often classified as black-box or post-hoc explainable. Predictive power is high, but interpretability demands specialized tools and techniques.

5 EMT: further remarks

The EMT algorithms presented in this study rely on data-driven and analytical methods when sufficient training data are available. This allows for the estimation of the mean and variance of each input coefficient through random sampling. To ensure optimal EMT performance, it is essential to use a complete, cleaned dataset and carefully select relevant input features. Techniques such as stepwise regression, principal component analysis, and correlation analysis can be employed as preprocessing steps prior to EMT model training to enhance predictive accuracy. The methodology for integrating feature engineering with EMT is detailed in a separate publication by Hasan and Lu (2023a). In situations where data are limited and random sampling is not feasible, domain experts with relevant process knowledge can provide reliable parameter estimates, including the most likely, minimum, and maximum values. In such cases, the mean can be approximated as a weighted average of the three estimates, and the standard deviation as one-sixth of the range, assuming a normal distribution.

By combining data-driven approaches with expert judgment, the hybrid EMT framework offers a flexible and practical solution for handling varying levels of data availability in predictive modeling. This integration improves prediction accuracy and reliability, supporting more informed decision-making and efficient resource allocation across industrial applications.

The comparative analysis establishes EMTs as the leading explainable AI approach for construction and broader civil engineering applications, successfully reconciling the often-competing demands of model interpretability and predictive accuracy. Where necessary, the framework permits calibrated reductions in accuracy to achieve essential transparency—a

strategic compromise that reinforces accountability while maintaining practitioner confidence through integrated variance estimation. This dual capability to generate both predictions and associated reliability metrics significantly enhances practical deployment potential.

Functionally, EMTs represent an advanced synthesis of decision tree architecture and regression analysis. The algorithm's internal nodes implement binary splits based on engineering parameters such as material specifications or environmental conditions, progressively refining data homogeneity. Terminal nodes then apply optimized linear regression models to these stratified subsets. This hierarchical structure preserves the intuitive logic of decision trees while incorporating the quantitative precision of regression analysis, offering complete traceability from input variables to final predictions. For critical engineering decisions affecting cost control, structural integrity, and environmental impact, this transparent causality proves indispensable. The regression components further enable precise sensitivity analysis, allowing engineers to numerically evaluate how parameter adjustments influence outcomes.

EMTs demonstrate particular advantages in civil engineering contexts compared to conventional machine learning approaches. Their explicit decision pathways document the exact contribution of each input variable, fostering trust in AI-assisted solutions. The adaptive partitioning mechanism captures complex, context-dependent relationships while maintaining computational efficiency. Perhaps most significantly, the self-documenting model architecture facilitates clear communication between technical teams, project stakeholders, and regulatory bodies.

However, EMT performance remains subject to universal machine learning constraints regarding data quality and completeness. Potential instability in predictions, reflected in elevated variance measures, can be mitigated through disciplined feature determination and dimensionality reduction during preprocessing. Successful implementation ultimately requires thoughtful adaptation to specific engineering subdomains, ensuring the algorithm's explainability capabilities align with disciplinary conventions and regulatory expectations. When properly configured, EMTs deliver a robust analytical framework that satisfies both technical requirements and organizational needs for verifiable, audit-ready decision support.

6 XAI application in construction engineering and opportunities for EMT

The civil engineering field has seen growing adoption of XAI techniques across diverse applications, from infrastructure modeling to environmental assessment. Researchers are increasingly combining machine learning with interpretation methods to enhance transparency in critical decision-making processes.

In infrastructure and environmental studies, Kwak and Lee (2024) applied eXtreme Gradient Boosting (XGBoost) with feature importance analysis and Accumulated Local Effects (ALE) to assess road transport's impact on groundwater quality. Similar approaches have proven valuable in structural engineering, where Gharagoz et al. (2025) used SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) to analyze seismic retrofitting performance predictions from XGBoost models.

Building energy forecasting represents another active area, with Leuthe et al. (2024) comparing transparent models (Linear Regression, Decision Trees) against ANNs, using Dependence Plots (PDP), ALE, LIME and SHAP for interpretation. Concrete strength prediction has similarly benefited from these techniques, as demonstrated by Guzmán-Torres et al. (2024) with DNNs and SHAP, and by Elhishi et al. (2023) finding XGBoost superior for this task when paired with SHAP explanations.

Emerging innovations include Component-Based Machine Learning (CBML) (Geyer et al., 2024) for energy-efficient design and BIM-integrated approaches (Reddy et al., 2025) for infrastructure maintenance. Transportation safety applications are also developing, such as Shuai et al.'s (2024) SHAP-interpreted SVM models for winter road conditions.

Geotechnical and geohazard applications show particular promise. Noureldin et al. (2025) successfully interpreted Gradient Boosting (GBoost) predictions of clay shear strength using SHAP and LIME, while Alqadhi et al. (2024) combined deep learning architectures with SHAP for landslide susceptibility mapping. Sustainable design benefits are evident in Tariq et al.'s (2025) ANN-based facade energy storage models and Khan et al.'s (2024) LightGBM framework for green buildings.

Structural assessment continues to advance through techniques like those of Shabbir et al. (2025), applying SHAP, PDP and ICE plots to retrofitting analyses. Smart infrastructure applications are emerging as well, exemplified by Dobrovolskis et al.'s (2023) XAI-enabled IoT systems for home energy management.

While post hoc explanation methods currently dominate, the field continues to grapple with balancing interpretability and performance. Developing ante hoc solutions that maintain both transparency and predictive power remains a key challenge for implementing trustworthy AI systems in civil engineering practice.

Despite the growing promise of XAI applications, a fundamental challenge remains: balancing predictive

accuracy with interpretability (Waqar, 2024b). While complex models such as Deep Neural Networks (DNNs) often deliver high predictive performance, they tend to function as opaque "black boxes," making their internal decision-making processes difficult to understand. To mitigate this limitation, post hoc interpretability techniques—such as SHAP, ALE, and LIME—have been widely adopted. However, EMTs offer a compelling alternative as an ante hoc XAI method that inherently integrates both accuracy and transparency (Hasan, 2023). EMT models can be particularly valuable in construction engineering predictive analytics, offering the following capabilities:

- Exploratory Power – Supports "what-if" scenario analysis to assess potential outcomes under varying project conditions.
- Control Process Explanation – Provides interpretable insights into how input variables influence decisions and highlights associated risks.
- Evidence-Based Decision-Making – Facilitates data-driven decision-making by enabling users to reference and analyze historical outcomes for similar input conditions.

7 Conclusion

This paper addresses ML algorithms commonly associated with providing AI based decision support in addressing complex problems in civil construction engineering. The relationships between inputs and outputs of a problem are generalized using ML algorithms on a training dataset, with the aim of gaining insights into the underlying problem and developing a prediction tool for analyzing new instances within the problem domain. XAI is increasingly crucial in civil engineering, a field where decisions impact safety, finances, and society. XAI aids in making AI operations transparent, which is essential for applications such as structural safety and regulatory compliance, where explaining AI reasoning processes is critical for establishing trust and accountability.

Through the literature review, this paper identified a set of attributes categorized by model type, explainability, model perspective, and interpretability—for evaluating the effectiveness of ML algorithms in terms of providing potential XAI decision support. Numerous construction engineering problems involve mapping input–output relationships to generate an equation or numerical model for approximating the relationships between the input space and the observable responses of the complex problem. Sufficient data is typically obtained from well-designed experiments carried out in physical labs or through virtual simulations to calibrate regression equations or AI model parameters, with the goal of minimizing deviations of the model-produced outputs from the target outputs in the training dataset. The EMT proposed by the authors is assessed against relevant XAI attributes as well as commonly used machine learning algorithms in construction engineering. Ultimately, the EMT shows potential in offering XAI solutions for applications in construction engineering and the broader civil engineering field.

Looking ahead in the civil engineering domain, AI and XAI need to be balanced to provide effective solutions, akin to the interaction of *Yin* and *Yang* based on ancient Chinese philosophy. AI models and algorithms continue to advance in speed and energy efficiency, for training large models while optimizing convergence performance and preventing overfitting. In parallel with the AI model, XAI serves as an approximate alternative model trained on the same dataset or a sample of a large dataset, with the aim of explaining how AI reaches its outcomes and understanding the reasoning logic of the AI model. This research focuses on the specific requirements of construction engineering applications for XAI, addressing the aspects of the problem, data, and model that necessitate explanation. The study seeks to formalize XAI frameworks tailored to applications in the civil construction field. Through enhanced transparency and logical reasoning, XAI will complement AI by fostering trust between AI and end users, thereby maximizing AI's potential.

Appendix A

EMT Prediction Performance.

The empirical performance evaluation, conducted in a separate study by Hasan and Lu (2023a), involved developing a productivity model using EMT and testing it against both MLR (Multiple Linear Regression) and Artificial Neural Network (ANN) models with the same inputs and dataset. The performance of these models is summarized in Table A1. Results show that the EMT model consistently outperforms the standalone MLR model and achieves accuracy comparable to that of ANN in terms of RMSE, depending on the dataset and application context. Determining whether EMT outperforms other models, such as ANN, requires careful experimentation aligned with specific data and application needs.

See Table 2

Author contributions
Both authors actively contributed to the development of the research and manuscript.

Table 2 Performance evaluation of various prediction models independently developed for labor productivity estimation in a case study by Hasan and Lu (2023a)

Performance Index	EMT		MLR Model		ANN Model	
	Training Set	Testing Set	Training Set	Testing Set	Training Set	Testing Set
R^2	0.93	0.97	0.88	0.87	0.92	0.89
Root mean square error (RMSE)	0.0139	0.0167	0.0191	0.0073	0.0169	0.0077
Average—variance index $(\alpha_{P,avg})$	0.29	0.25	–	–	–	–

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Data availability
All materials and publications reviewed in this study are publicly available online.

Declarations

Competing interests
The authors declare no competing interests relevant to this article.

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